

## **Tutorial 8(24/9/26)**

1. **Hospital Emergency Monitoring System:** A hospital monitoring application uses three threads: EmergencyAlert for critical patient alerts, VitalMonitor for continuously checking vital signs, and ReportGenerator for preparing routine reports. Develop a Java multithreaded program that assigns the highest priority to EmergencyAlert, a medium priority to VitalMonitor, and the lowest priority to ReportGenerator. Display the name and priority of each thread during execution using getName() and getPriority().
2. **An online food delivery application** performs multiple tasks concurrently, such as processing customer orders, tracking delivery locations, and sending order-status notifications. Develop a Java program using multithreading by creating three separate threads named OrderProcessing, DeliveryTracking, and Notification. Assign different priorities to the three threads based on the importance of their tasks using setPriority(). Start all the threads from the main method and display the thread name, assigned priority, and activity being performed to demonstrate multiple thread creation, thread priority, and concurrent execution.
3. **Student Registration System:** Develop a Java Swing application for student registration using JFrame. Provide JTextField components to enter the student name and register number, JRadioButton components to select gender, and JComboBox to select the department. Include a JButton to submit the details and display the entered information using a JOptionPane.
4. **User Login and Preferences:** Develop a Java Swing application for a user login system. Use JTextField to enter the username and JPasswordField to enter the password. Provide JCheckBox components for selecting user preferences such as Remember Me and Receive Notifications. On clicking the Login button, display an appropriate login message using JOptionPane.
5. **Student Course Management System:** Design a Java Swing application to maintain student course information. Use a JList to display the available courses and a JTable to display the student name, selected course, and enrollment status. Provide suitable JButton components to add and remove course registrations and use a JScrollPane wherever required to handle larger amounts of information.
6. **Text Editor Application:** Develop a simple Java Swing-based text editor using JTextArea for entering and editing text. Provide a JMenuBar containing a File menu with JMenuItem options such as New, Clear, and Exit, and an Edit menu with appropriate editing options. Place the text area inside a JScrollPane to allow scrolling when the text exceeds the visible area.